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Graph neural networks for identifynding robust biomarkers in multiomics data

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Task

- disease subtype classification
- survival prediction
- drug response prediction
- we want explainable models, that can hint at the underlying biomarkers

Multi-omic data includes

- expression data
    - mRNA, miRNA, circRNA
- DNA methylation
- Copy number variations (CNV)
- TE counts

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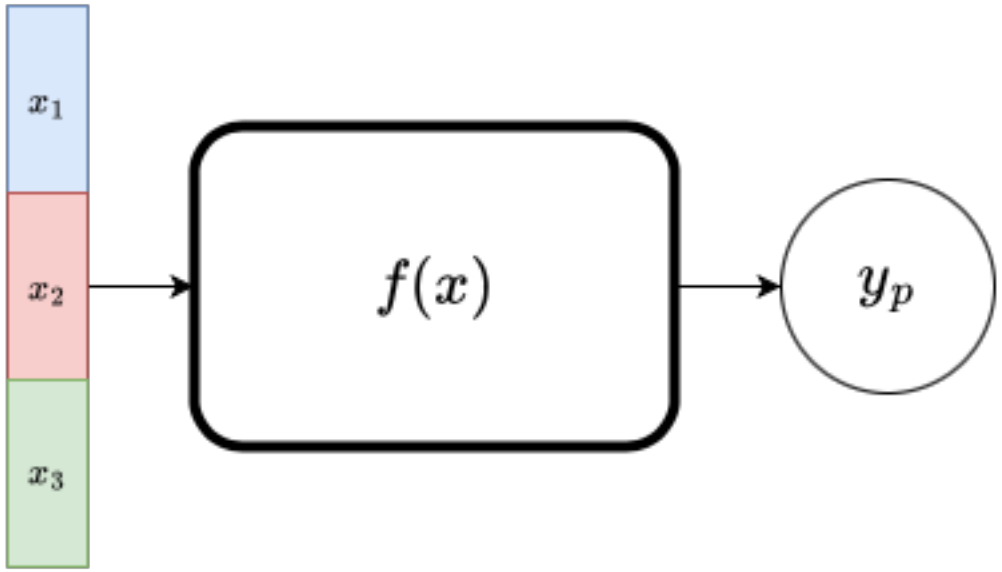
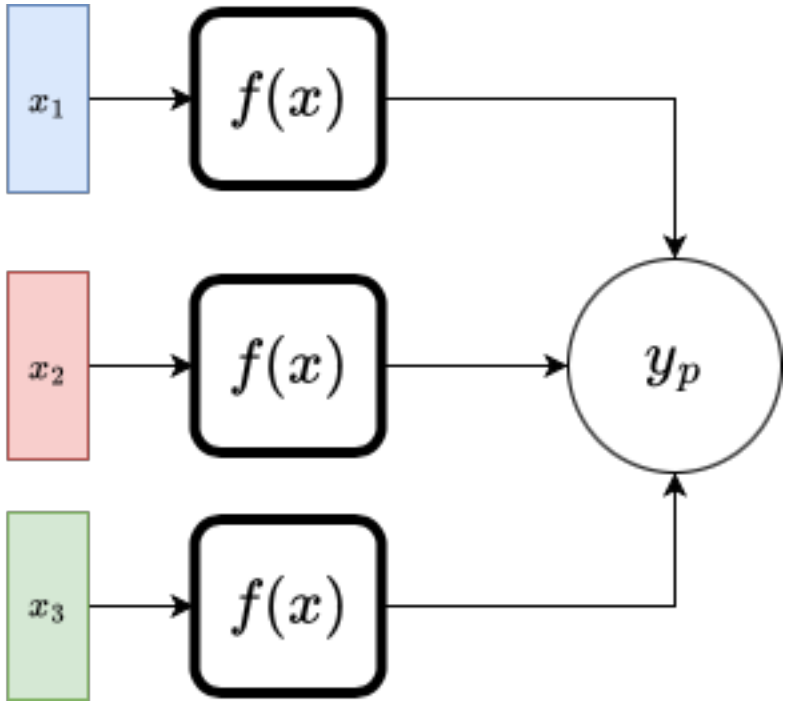
Datasets

- Breast invasive carcinoma (BRCA)
- Myelodysplastic syndromes (MDS)
- Others, for future work
    - Liver cancer (KIPAN)
    - Lower grade glioma (LGG)
    - Alzheimer disease (ROSMAP)

Integrating data





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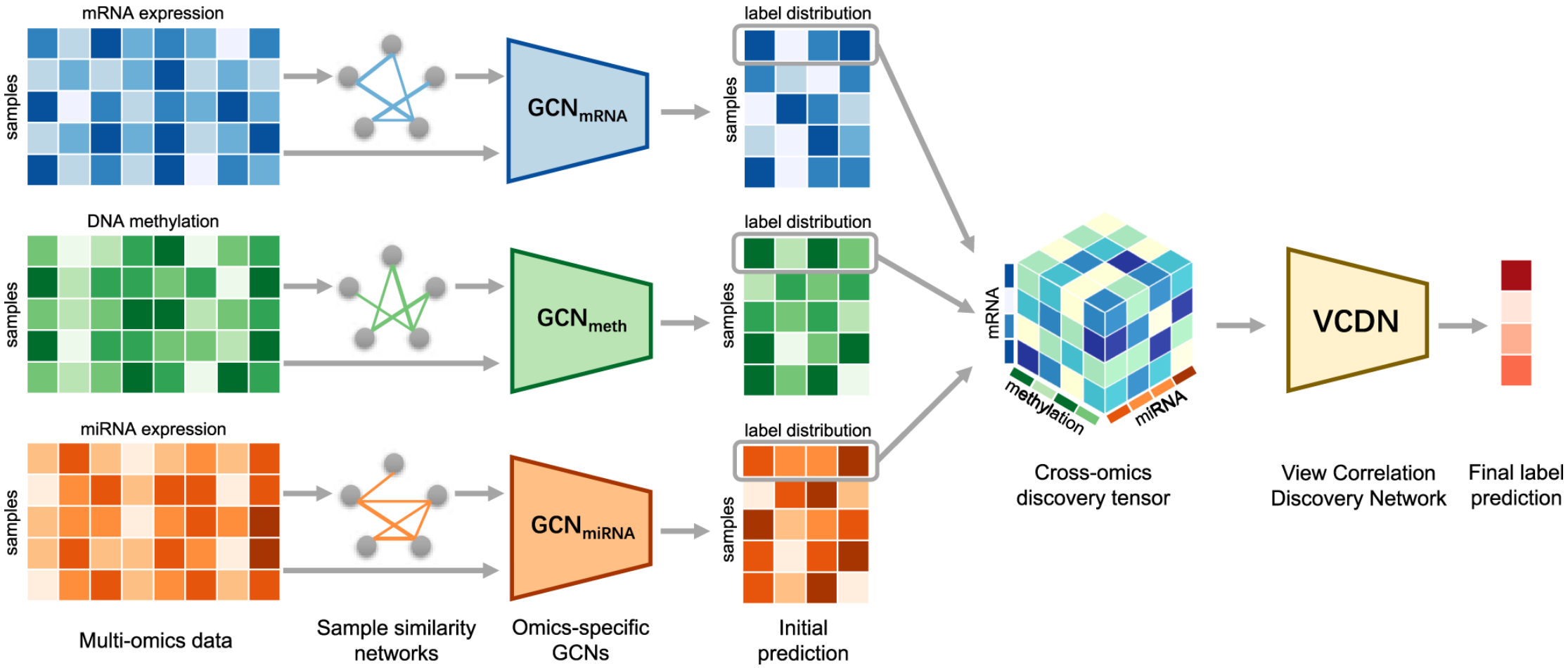
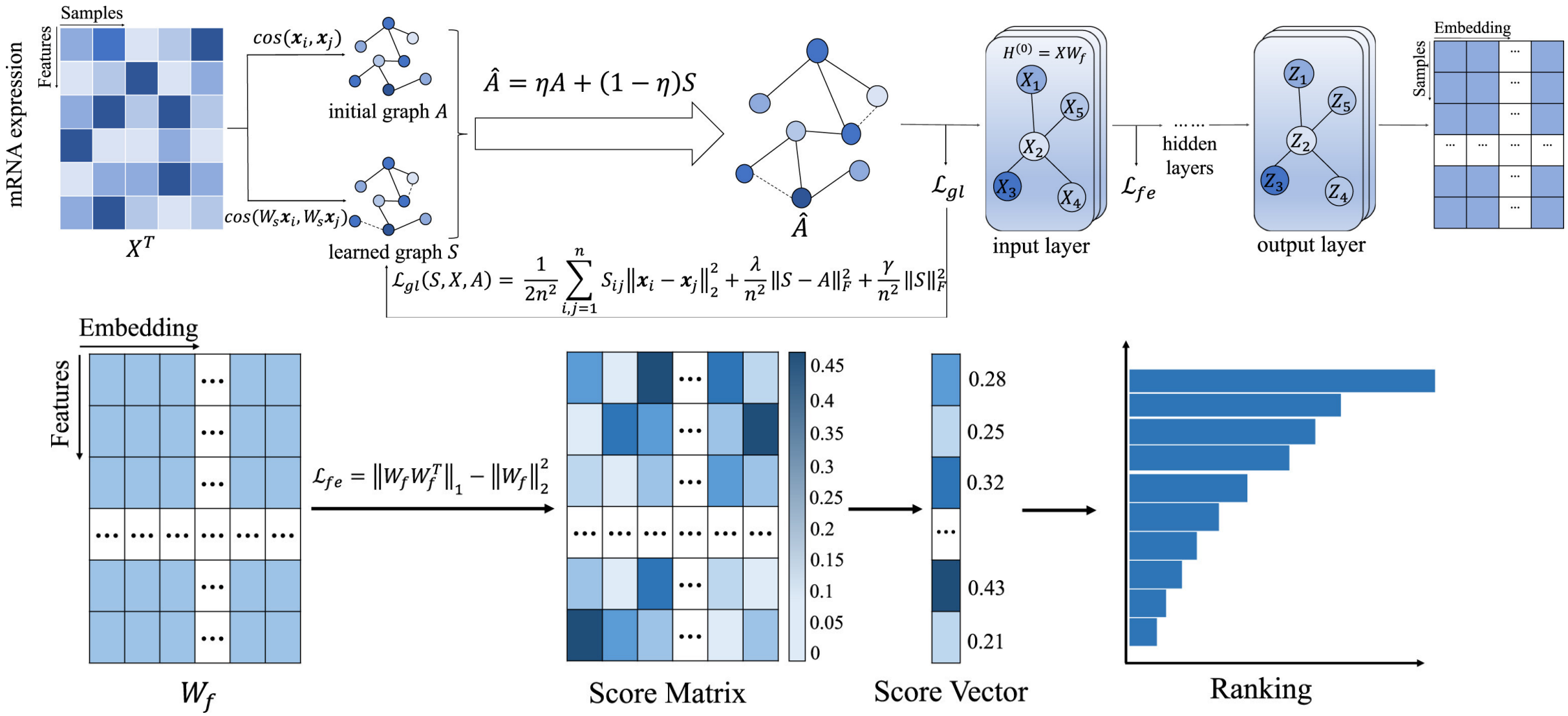
Data preprocessing

- normalize reads with RPKM, FPKM or TPM
- log transform with $y = \log_2(x + 1)$
- filter out uninformational features
- normalize to zero mean and unit variance

Mogonet


mRNA expression
samples
DNA methylation
samples
miRNA expression
samples
GCN$_{mRNA}$
GCN$_{meth}$
GCN$_{miRNA}$
label distribution
samples
label distribution
samples
label distribution
samples
mRNA
methylation
miRNA
VCDN
Multi-omics data
Sample similarity networks
Omics-specific GCNs
Initial prediction
Cross-omics discovery tensor
View Correlation Discovery Network
Final label prediction

Moglam


Samples
Features
mRNA expression
X^T
$cos(\boldsymbol{x}_i, \boldsymbol{x}_j)$
initial graph A
$cos(W_s\boldsymbol{x}_i, W_s\boldsymbol{x}_j)$
learned graph S
$\hat{A} = \eta A + (1-\eta)S$
$\hat{A}$
$\mathcal{L}_{gl}$
$H^{(0)} = XW_f$
X_1 X_2 X_3 X_4 X_5
input layer
$\mathcal{L}_{fe}$
hidden layers
Z_1 Z_2 Z_3 Z_4 Z_5
output layer
Embedding
Samples
$\mathcal{L}_{gl}(S,X,A) = \frac{1}{2n^2}\sum_{i,j=1}^{n} S_{ij}\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2 + \frac{\lambda}{n^2}\|S-A\|_F^2 + \frac{\gamma}{n^2}\|S\|_F^2$
Embedding
Features
W_f
$\mathcal{L}_{fe} = \|W_f W_f^T\|_1 - \|W_f\|_2^2$
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
Score Matrix
0.28
0.25
0.32
0.43
0.21
Score Vector
Ranking

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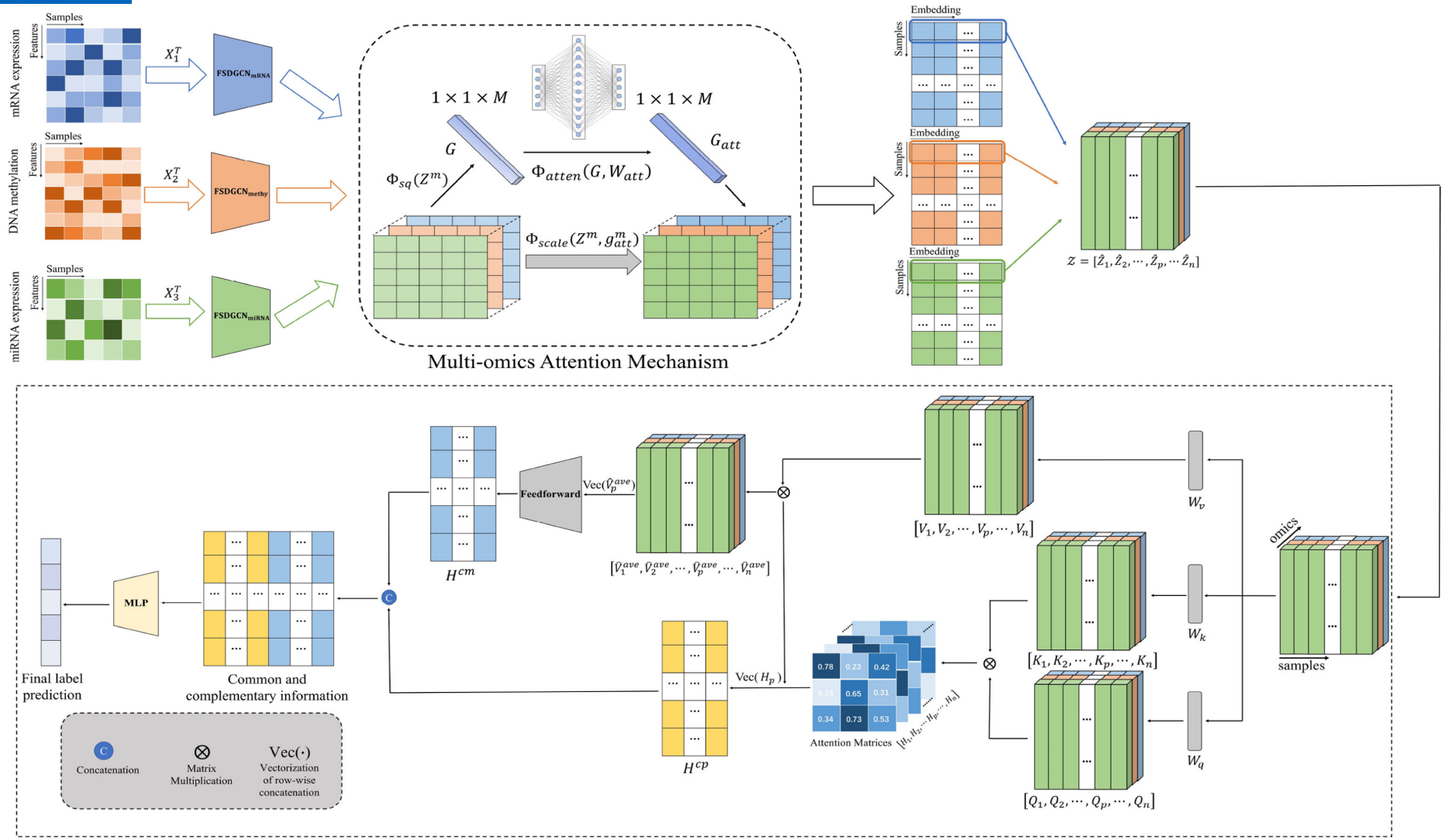

Omic-integrated Representation Learning

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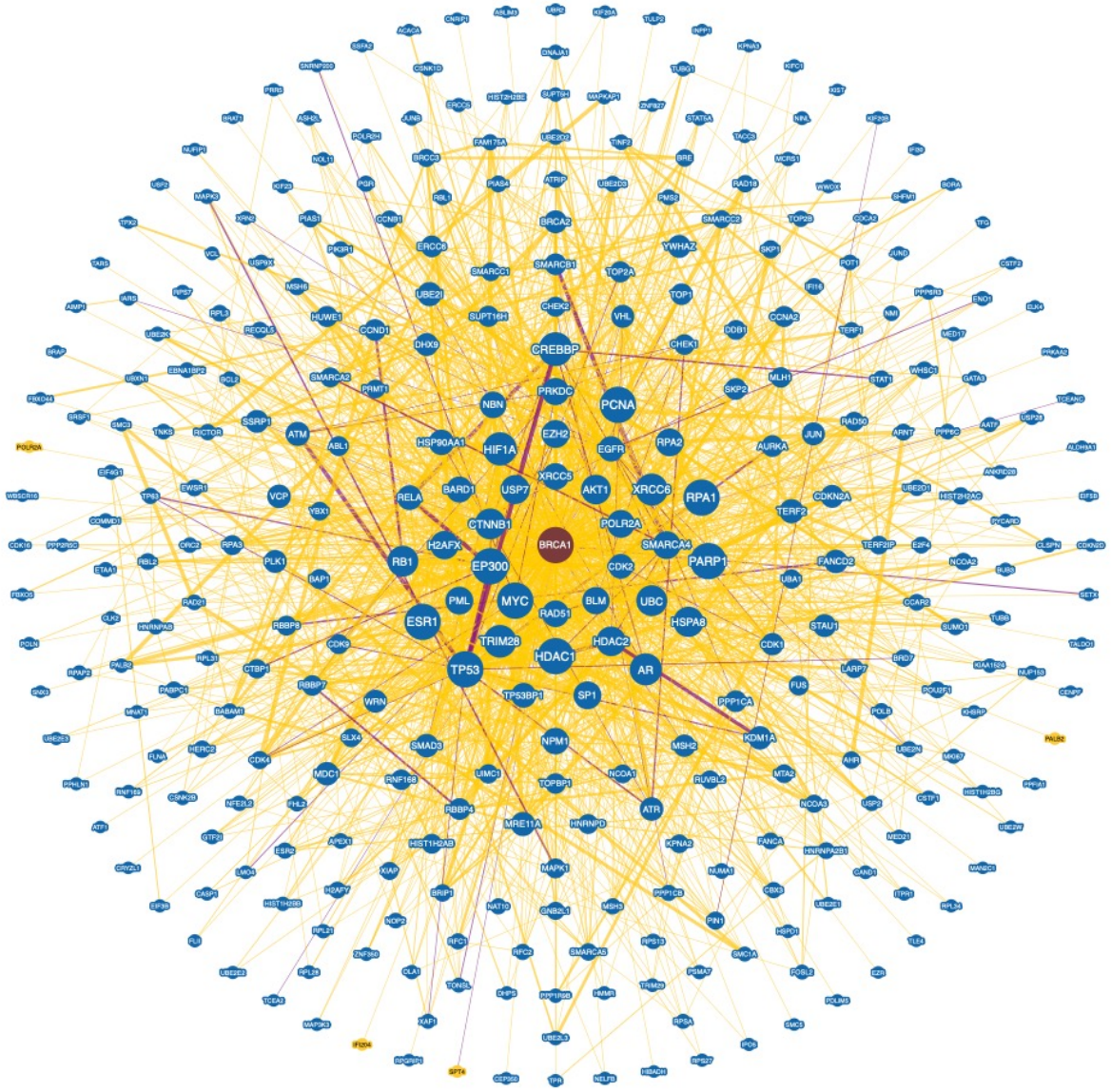
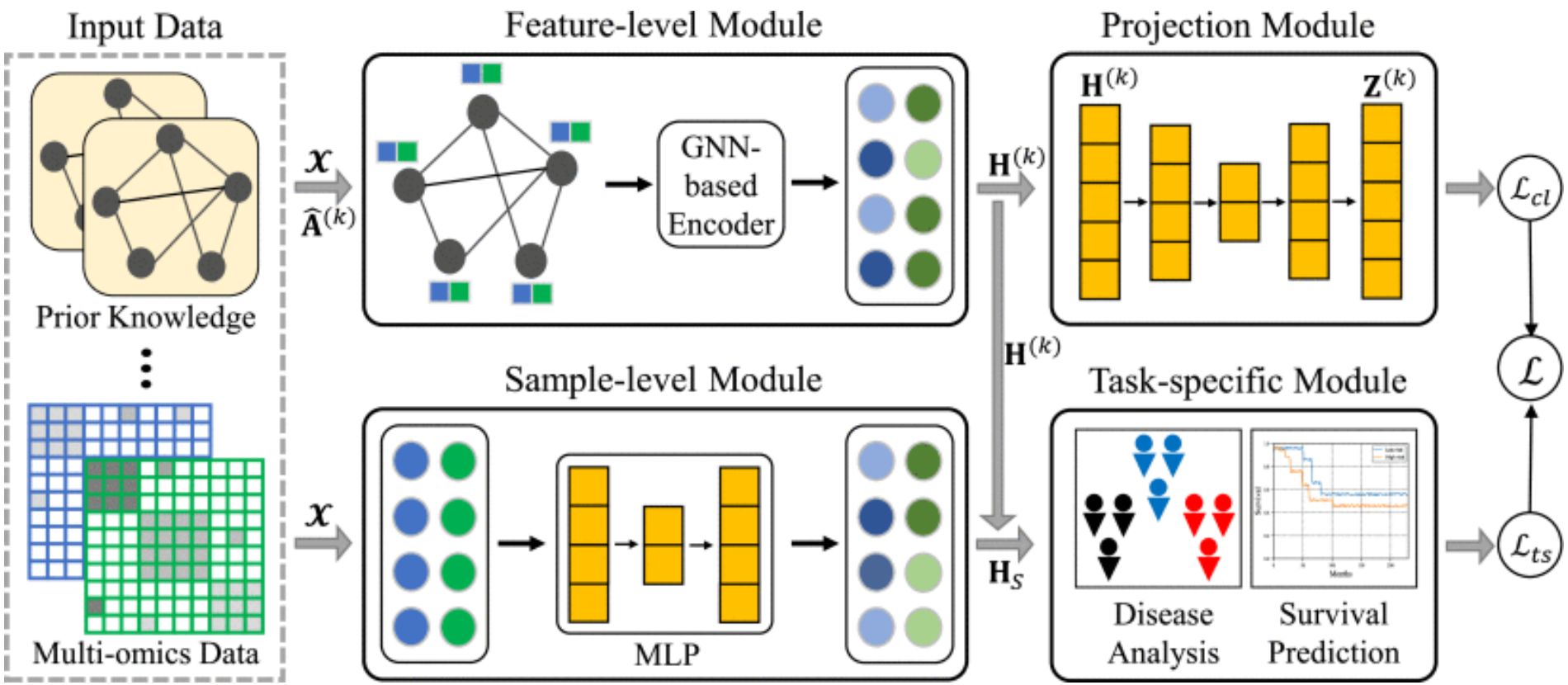
Feature interactions

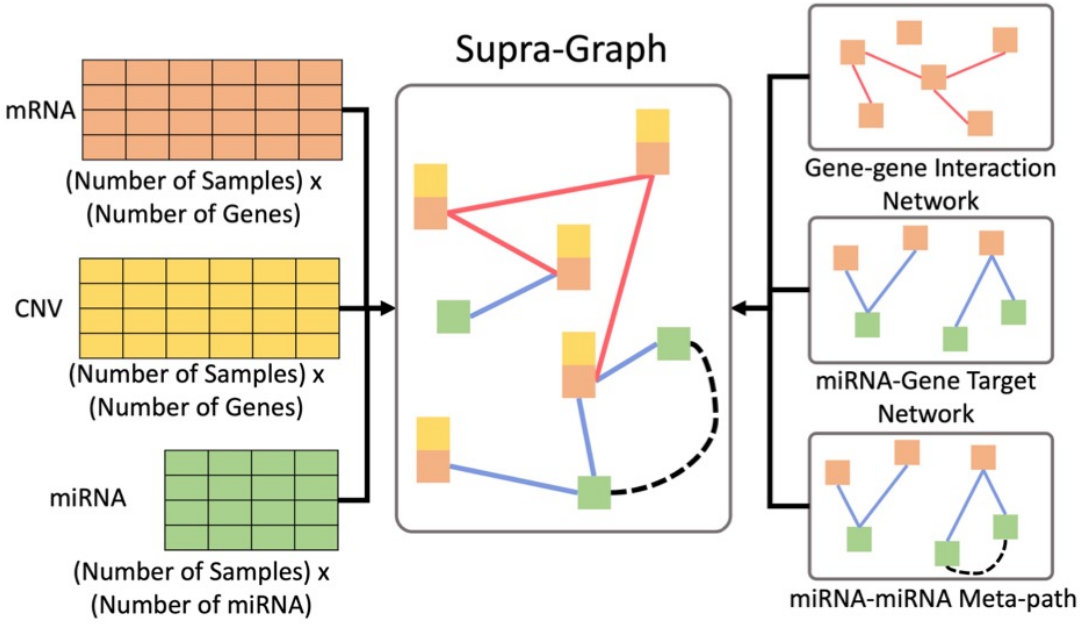
- gene - gene interactions
- protein - protein interactions
- miRNA - gene interactions

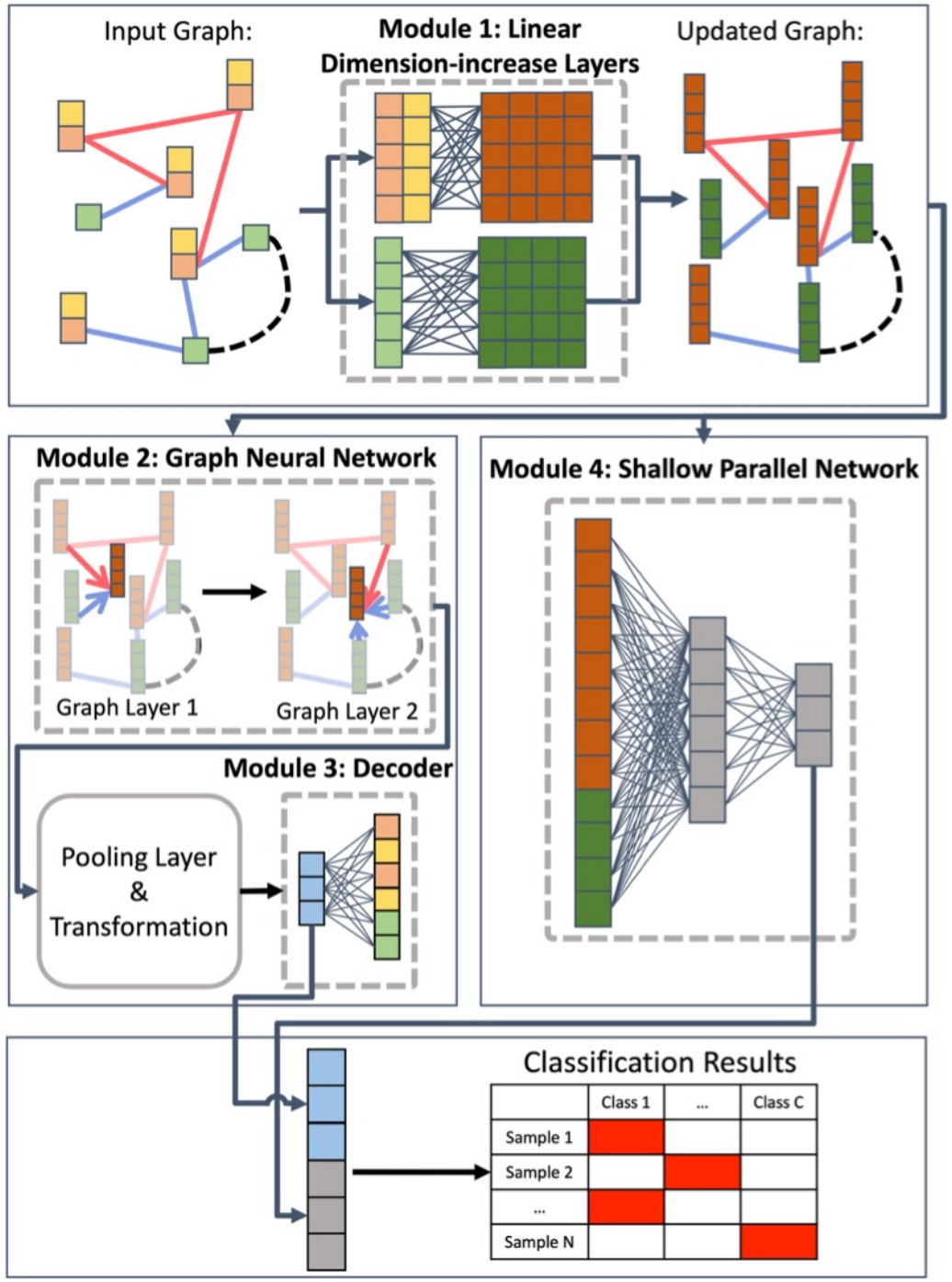


MPK-GNN


Input Data
Prior Knowledge
Multi-omics Data
Feature-level Module
GNN-based Encoder
Sample-level Module
MLP
Projection Module
Task-specific Module
Disease Analysis
Survival Prediction

Li and Nabavi





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Performance comparision

Method	Accuracy	F1 macro	F1 weighted
Linear SVM	0.729 ± 0.018	0.64 ± 0.017	0.702 ± 0.017
KNN	0.74 ± 0.02	0.67 ± 0.02	0.72 ± 0.02
XGBoost	0.771 ± 0.008	0.754 ± 0.01	0.701 ± 0.17
Mogonet	0.7886 ± 0.021	0.7740 ± 0.029	0.7254 ± 0.037
MPK-GNN (2-omics only)	0.7742 ± 0.034	0.7821 ± 0.031	0.7365 ± 0.042
MOGLAM	0.8380 ± 0.023	0.8456 ± 0.022	0.8124 ± 0.028
Li and Nabavi	0.864	0.875	-

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What is next

- Expand the review, go into detail and find which methods work and which do not
- Provide a validation across several datasets for all the methods
- Utilize ideas from different methods and come up with one what will work better and find new biomarkers

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Thank you for your $softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V$